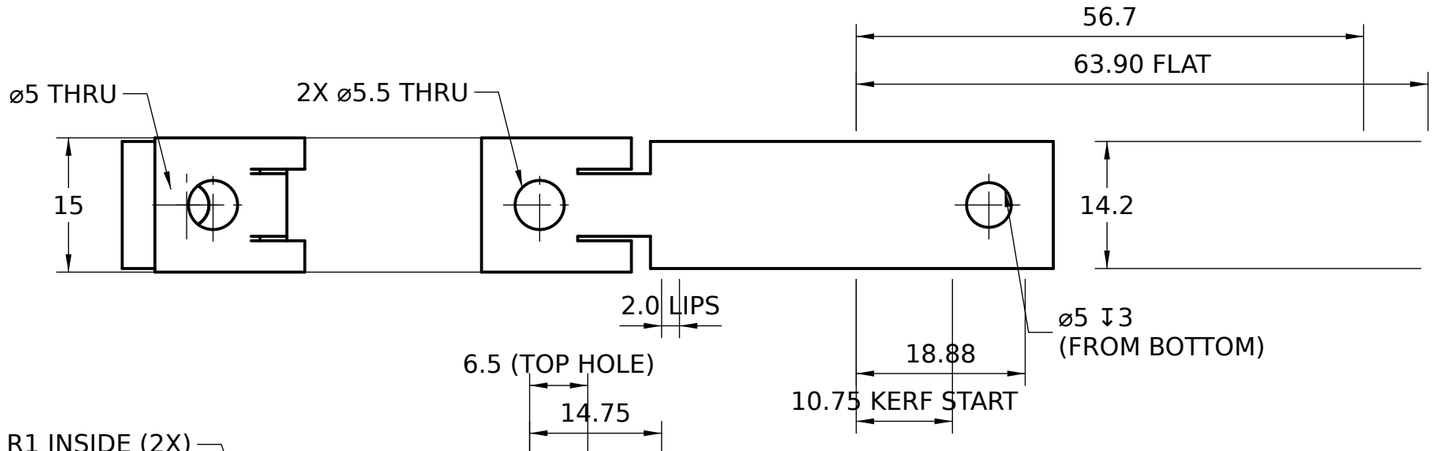
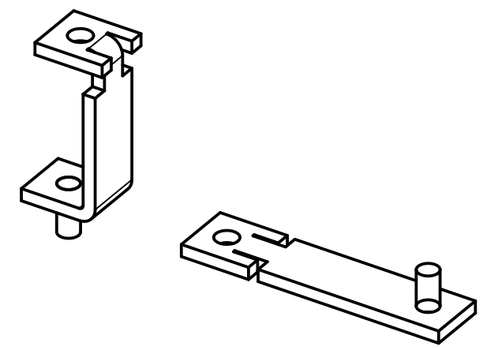
REVISIONS			
REV	DESCRIPTION	BY	DATE
A	Initial	Claude	17/10/26
B	The 2 PEM CLSS-M5-2 nuts are in the model (heads inside the C, on the formed strap and the flat blank); new stages model: nuts beside their holes, both pressed with the 0.6 trim piece as its own body, trimmed	Claude	18/10/26
C	Lug PEM 7.1 from the inside bend (N36 = A, R198): lug hole and bottom leg end 3.35 further from the web; blank 63.7 (cut 64.3)	Claude	18/10/26
D	N37: lug end = PEM FHS-M5-12 flush stud (hole 5.0, 7.2 from the bend and the leg end); top end = 5.5 clearance hole for the ear bolt into a DIN 934 nut in a printed holder; no PEM nuts, nothing trimmed; blank 63.9	Claude	19/10/26





TRIMETRIC VIEW (SCALE 1:1.25)

NOTES:

1. C110 TINNED COPPER FROM 3 x 15 BAR (BOPAODAO B09Z7NBLDK). 14 PER CAR. KEEP THE TIN ON THE CLAMP FACES.
2. SAW 63.90 LONG. DRILL ⌀5.5 AT THE TOP HOLE, ⌀5.0 +0.08/0 AT THE LUG HOLE.
3. SLOTS 0.5 WIDE, 3.5 - 4.0 EACH SIDE OF CENTRE, 10.75 TO 16.75; NOTCH 3.5 - 7.5 TO 18.88; FILE TO 14.2 WIDE FROM 18.88 TO THE END.
4. PRESS THE PEM FHS-M5-12 STUD IN FROM THE INSIDE FACE UNTIL ITS HEAD IS FLUSH. NOTHING IS TRIMMED.
5. BEND 2 x 90 DEG, R1.0 INSIDE, AT 12.31 AND 47.94 FROM THE TOP END; BEND 1 ON THE TONGUE ONLY. CHECK THE FIRST PART.
6. FIRST PART: M5 FLANGE NUT ON THE STUD AT 6 N·m; THE STUD MUST NOT TURN OR PUSH OUT. SCRAP THAT PART. ASSEMBLY: SHEET 2.

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
POWER OUTPUT C-STRAP 3 x 15 (FORMED + FLAT BLANK)

PART / DWG NO.	REV	MATERIAL
ST185-FB-117	**D.01**	C110 COPPER, TINNED

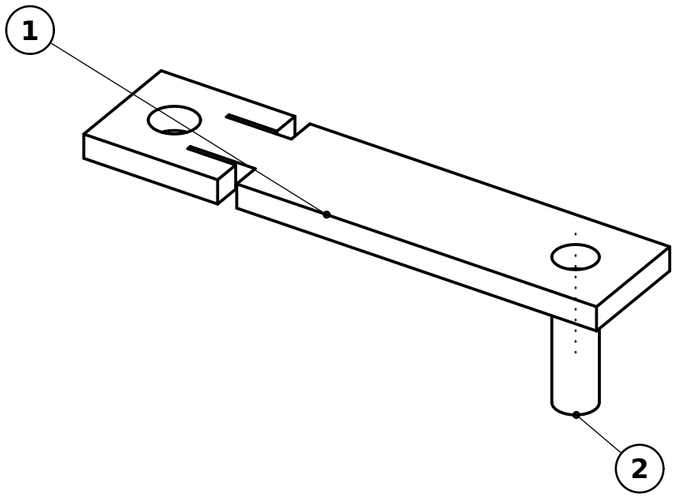
SCALE	UNITS	SIZE	SHEET	DATE	DRAWN
1.5:1	mm	B (11x17)	1 OF 2	2026-10-19	SteveCAD

SOURCE
Power_Strap_Out_CuC110_15x3_C.step (Fusion) + flat blank from spec.strap_blank() (build123d)

3RD ANGLE

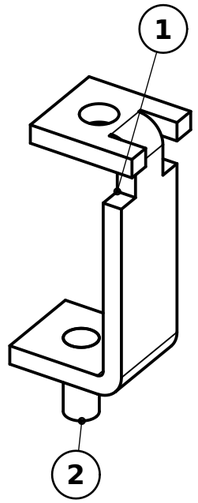
STEP 1: PRESS THE STUD IN FROM THE INSIDE FACE

FLAT BLANK, EXPLODED (NOT TO SCALE)



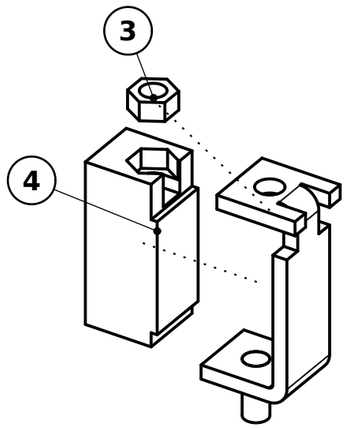
STEP 2: STUD FLUSH, THEN BEND 2 x 90 DEG

NOTHING IS TRIMMED (NOT TO SCALE)



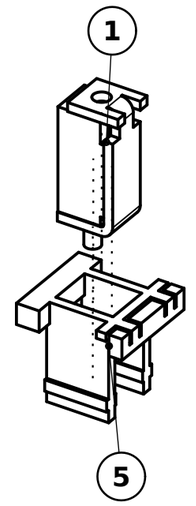
STEP 3: NUT INTO THE HOLDER, HOLDER INTO THE C

EXPLODED (NOT TO SCALE)



STEP 4: DROP THE BUILT STRAP STRAIGHT DOWN INTO ITS WELL

BASE CUT DOWN TO ONE WELL (NOT TO SCALE)



UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN MILLIMETERS
TOLERANCES:
X ±0.5 X.X ±0.2
X.XX ±0.05 ANGLES ±0.5°
SURFACE FINISH Ra 3.2 µm
INTERPRET PER ASME Y14.5-2018

COMPANY / PROJECT
ST185 CELICA GT-FOUR - LINK ECU CONVERSION

TITLE
POWER OUTPUT C-STRAP 3 x 15 (FORMED + FLAT BLANK)

PART / DWG NO.	REV	MATERIAL
ST185-FB-117	**D.01**	C110 COPPER, TINNED

SCALE	UNITS	SIZE	SHEET	DATE	DRAWN
AS SHOWN	mm	B (11x17)	2 OF 2	2026-10-19	SteveCAD

SOURCE
Power_Strap_Out_CuC110_15x3_C.step (Fusion) + flat blank from spec.strap_blank() (build123d)

3RD ANGLE